Reference FASTA

Scan contigs and choose padded layout

Encode forward genome

Append reverse complement

Count suffix prefixes and plan bins

Sort suffixes within prefix bins

Pack suffix-array coordinates

Build SAindex prefix ranges

Optional GTF / splice junctions

Prepare junction sequences

Junctions supplied?

Yes

Search junction suffix insertion positions

Sort and merge inserted suffixes

Pack augmented suffix array

Rebuild SAindex

No

Write complete index

Genome, SA, SAindex, chromosome and annotation tables